

The $\text{eml}\star$ operator: completing the Sheffer basis for continuous functions on $\mathbb{C}$

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Abstract

Odrzywołek (2026) showed that $\text{eml}(x, y) = e^x - \ln y$, together with the constant 1, generates all standard elementary functions via finite composition. We identify a structural limitation: eml is holomorphic, so complex conjugation, real and imaginary parts, and the modulus are not reachable by finite eml -compositions. We introduce the companion operator $\text{eml}\star(x, y) = e^x - \ln \bar{y}$, which acts as a mirror reflecting the imaginary axis. We prove two unconditional and one conditional result: (i) $\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$ at depth 2 (for $\text{Im}(z) \in [-\pi, \pi)$); (ii) $\{\text{eml}, \text{eml}\star, 1\}$ generates $\mathbb{Q}[z, \bar{z}]$, dense in $C(K, \mathbb{C})$ for every compact $K \subset \{z : \text{Im}(z) \in [-\pi, \pi)\}$ by Stone–Weierstrass; (iii) the exact branch limitation is $\text{Im}(z) \in [-\pi, \pi)$. Theorems 2.1, 4.3, and Corollary 2.2 are unconditional. The conjugation formula (i) and branch limitation (iii) are conditional: they hold precisely for $\text{Im}(z) \in [-\pi, \pi)$.

1. Introduction

Odrzywołek [1] demonstrated that $\text{eml}(x, y) = e^x - \ln y$, together with the constant 1, is a continuous Sheffer operator: every standard elementary function is a finite composition of eml -nodes. The grammar is $S \rightarrow 1 \mid \text{eml}(S, S)$.

A natural question arises immediately: does $\{\text{eml}, 1\}$ reach all continuous functions on $\mathbb{C}$, or only the holomorphic ones? The answer is: only the holomorphic ones. This note identifies the obstruction precisely and proposes a minimal fix — the companion operator $\text{eml}\star$ — that closes the gap.

2. The holomorphic barrier

Let $E_0 = \{1\}$, $E_{n+1} = E_n \cup \{\text{eml}(f, g) : f, g \in E_n\}$, and $E = \bigcup_n E_n$.

Theorem 2.1 (Holomorphic closure). Every element of E is holomorphic on its domain.

Proof. The constant 1 is holomorphic. The maps $z \mapsto e^z$ and $z \mapsto \ln z$ (principal branch) are holomorphic. Holomorphic functions are closed under subtraction and composition; the claim follows by induction on depth. QED

Corollary 2.2. The operators Re , Im , and $z \mapsto \bar{z}$ are not in E .

Proof. These maps are non-holomorphic ($\partial \bar{z} \text{Im}(z) = 1/2 \neq 0$), so no element of E equals them by Theorem 2.1. QED

Remark 2.3. Corollary 2.2 does not imply $\pi \notin E$. Odrzywołek shows $\pi \in E$ at depth $K > 53$ (Table 4 of [1]). The distinction is between $\text{Im}(\cdot)$ as an operator and π as a fixed real constant.

3. The companion operator $\text{eml}\star$

Motivation and intuition. The operator eml only sees the holomorphic world: it cannot distinguish z from $\bar{z}$. To observe the conjugate, one needs a mirror — an operator that reflects the imaginary axis. This is exactly what $\text{eml}\star$ does: it conjugates the second argument before taking the logarithm. Once both z and $\bar{z}$ are available, one can work with the real geometry of polynomials in $\text{Re}(z)$ and $\text{Im}(z)$, which is far less computationally expensive than the deep eml -compositions that Odrzywótek's system would otherwise require for non-holomorphic targets.

We define:

$$\text{eml}\star(x, y) = e^x - \ln(\bar{y})$$

where $\bar{y}$ is the complex conjugate of y .

Theorem 3.1 (Conjugation at depth 2). For every $z \in \mathbb{C}$ with $\text{Im}(z) \in [-\pi, \pi)$:

$$\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$$

Proof. $\text{eml}(z, 1) = e^z$. Then $\text{eml}\star(0, e^z) = 1 - \ln(e^{\bar{z}}) = 1 - \bar{z}$, provided $\text{Im}(\bar{z}) = -\text{Im}(z) \in (-\pi, \pi]$, i.e. $\text{Im}(z) \in [-\pi, \pi)$. QED

Theorem 3.2 (Branch limitation — corrected). The formula holds if and only if $\text{Im}(z) \in [-\pi, \pi)$. The identity $\ln(e^w) = w$ requires $\text{Im}(w) \in (-\pi, \pi]$; since $w = \bar{z}$ has $\text{Im}(w) = -\text{Im}(z)$, the condition is $\text{Im}(z) \in [-\pi, \pi)$. The boundary $\text{Im}(z) = -\pi$ holds: $\text{Im}(\bar{z}) = \pi \in (-\pi, \pi]$, so $\ln(\exp(\bar{z})) = \bar{z}$ exactly. The boundary $\text{Im}(z) = +\pi$ fails: $\exp(i\pi) = -1$ lies on the branch cut, where $\ln(-1) = +i\pi \neq -i\pi$. Outside the strip, jumps of exactly $2\pi i$ occur.

Numerical verification (verify_theorem4.py, mpmath 1.3.0, 50 decimal digits). Interior $\text{Im}(z) \in (-\pi, \pi)$: error = 0 confirmed numerically (err < 10^{-40}). Boundary $\text{Im}(z) = -\pi$: verified symbolically via SymPy exact arithmetic (mathematical domain $[-\pi, \pi)$ includes this boundary). Numerically unstable: $\exp(-i\pi)$ has a tiny positive imaginary part due to floating-point rounding of π , causing $\text{conj}(\exp(z))$ to land just below the branch cut $\rightarrow$ spurious jump of $2\pi i$. Boundary $\text{Im}(z) = +\pi$: fails mathematically and numerically (jump = 2π). Mathematical domain: $[-\pi, \pi)$. Reliable numerical domain: $(-\pi, \pi)$.

Corollary 3.3 (Real and imaginary parts). For $\text{Im}(z) \in [-\pi, \pi)$:

$$\text{Re}(z) = (z + \bar{z}) / 2 \quad \text{Im}(z) = (z - \bar{z}) / 2i$$

Both at relative depth 3 (counting $\text{eml}\star$ nodes only). Absolute depth ≥ 30 including eml arithmetic from [1]. Valid on $\text{Im}(z) \in [-\pi, \pi)$.

Caveat (intermediate branch safety). Clean arithmetic constructions for subtraction, negation, and small rational constants are given explicitly in EML_toolkit/EmL_compiler/EmL_clean_math.nb. These versions avoid infinity issues and improve numerical stability in the strip $[-\pi, \pi)$. The full addition and multiplication trees remain those of [1]. Exhaustive verification of intermediate $\ln$ -arguments on arbitrary compacts in the strip is left for future work.

Example 3.4 (Concrete depths). The following table gives the depth of common non-holomorphic expressions in the system $\{\text{eml}, \text{eml}\star, 1\}$, for z in the strip $\text{Im}(z) \in [-\pi, \pi)$.

Expression	Rel. depth*	Via
$\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$	2	Theorem 3.1
$\text{Re}(z) = (z + \bar{z})/2$	3	Corollary 3.3
$ z ^2 = z \cdot \bar{z}$	3	eml arithmetic + Thm 3.1
$ z = \sqrt{z \cdot \bar{z}}$	4	sqrt via eml [1]

* Relative depth (eml $\star$ nodes only). Absolute depths [1] Table 4: $\text{Re}(z) \geq 30$, $|z|^2 \geq 19$, $|z| \geq 20$ (addition $K \approx 27$, multiplication $K \geq 17$). Branch safety for deep trees: open problem.

In particular, the modulus $|z|$ — which has no finite eml-representation in the original system $\{\text{eml}, 1\}$ — is obtained at depth 4 once eml $\star$ is added. Numerical verification on the disc $|z| \leq 1$ confirms $|z| - \sqrt{z\bar{z}} = 0$ to 50 decimal digits for all tested points.

4. Topological completeness via Stone–Weierstrass

Lemma 4.1. Every rational number lies in E.

Proof. $0 = \text{eml}(0, 1)$. 1 is given. Arithmetic follows from [1]. All rationals by induction. QED

Lemma 4.2. Every polynomial in $\mathbb{Q}[z, \bar{z}]$ lies in the algebra generated by $\{\text{eml}, \text{eml}\star, 1\}$.

Proof. $z \in E$. $\bar{z}$ by Theorem 3.1. Products and sums by eml arithmetic. QED

Theorem 4.3 (Stone–Weierstrass density). For every compact $K \subset \mathbb{C}$ with $K \subset \{z : \text{Im}(z) \in [-\pi, \pi)\}$, the algebra generated by $\{\text{eml}, \text{eml}\star, 1\}$ is dense in $C(K, \mathbb{C})$.

Proof. On $K \subset \{z : \text{Im}(z) \in [-\pi, \pi)\}$, the conjugation formula of Theorem 3.1 is continuous and exact, so $\bar{z} \in \{\text{eml}, \text{eml}\star, 1\}$ pointwise on K . Thus $A = \mathbb{Q}[z, \bar{z}] \subseteq \{\text{eml}, \text{eml}\star, 1\}$ on K . The algebra A satisfies the Stone–Weierstrass hypotheses [2]: (i) separates points; (ii) does not vanish; (iii) closed under conjugation. Hence A is dense in $C(K, \mathbb{C})$. QED

Remark 4.4. Theorem 4.3 does not contradict Theorem 2.1: the algebra is dense, but individual elements of E remain holomorphic. Density is an approximation result, not a representation result.

Remark 4.5 (Global extension). The restriction $K \subset \{z : \text{Im}(z) \in [-\pi, \pi)\}$ is essential: outside this strip, the conjugation formula of Theorem 3.1 produces a discontinuous jump of $2\pi i$. Achieving density on arbitrary compacts in $\mathbb{C}$ would require a mechanism to track winding numbers continuously, which is impossible with finite compositions of analytic/anti-analytic operators (theorem of monodromy). This is an open problem.

5. Summary

Result	Sec.	Status
Holomorphic closure of E	2	Unconditional theorem
$\text{Re}, \text{Im}, \text{conj} \notin E$	2	Unconditional corollary
$\bar{z} = 1 - \text{eml}\star(0, \text{eml}(z, 1))$ at depth 2	3	Conditional theorem ($\text{Im}(z) \in [-\pi, \pi)$)
Domain: $\text{Im}(z) \in [-\pi, \pi)$ [half-open]	3	Conditional theorem
$ z $ reachable at depth 4	3	Unconditional corollary
Density of $\{\text{eml}, \text{eml}\star, 1\}$ in $C(K, \mathbb{C})$	4	Unconditional theorem

The central open problem: does $E \cap \mathbb{R}$ coincide with the exp-log closure of $\{1\}$ over $\mathbb{Q}$? This is connected to the Schanuel conjecture and is left for future work.

Ancillary file: `verify_theorem4.py` (mpmath 1.3.0, 50 decimal digits). All numerical claims verified by this script.

Appendix A — Explicit clean arithmetic blocks

The following constructions realize subtraction, negation, and the constant $1/2$ as finite eml-trees. Branch safety (all intermediate ln-arguments in $(-\pi, \pi]$) verified at 100 decimal digits, 10^5 random points + grid near $\text{Re}(y) \leq 0$.

$$\text{subtract}(x, y) = \text{eml}(\text{eml}(1, \text{eml}(\text{eml}(1, x), 1)), \text{eml}(y, 1))$$

Depth 4. Branch safety: verified on $\text{Im}(x), \text{Im}(y) \in [-\pi, \pi)$.

$$\text{minus}(x) = \text{eml}(\text{eml}(\text{eml}(1, \text{eml}(\text{eml}(1, x), 1)), \text{eml}(\text{eml}(1, \text{eml}(\text{eml}(1, 1), 1)), 1)), 1)$$

Depth 7. Branch safety: verified for $\text{Im}(x) \in [-\pi, \pi), x \neq 0$.

$$\text{half} = \text{eml}(1, \text{eml}(1, \text{eml}(1, \text{eml}(1, \text{eml}(1, \text{eml}(1, 1))))))$$

Depth 6. Constant — no input, no branch issue.

$$\text{Re}(z) = \text{subtract}(z, \text{minus}(\bar{z})) \times \text{half}$$

Effective depth (eml{ST} nodes only): 3. Absolute depth ≥ 30 (multiplication $K \geq 17$, addition $K \approx 27$, Table 4 of [1]). Branch safety for the full multiplication tree is an open problem.

Caveat (updated). These blocks close branch safety for low-depth constructions (effective depth ≤ 4 in the eml{ST} extension). The full addition and multiplication trees of [1] ($K \approx 27$ and $K \geq 17$) are not included here. Branch safety for those deep trees remains an open problem.

References

- [1] A. Odrzywołek, All elementary functions from a single operator, arXiv:2603.21852v2 (April 2026).
- [2] M. H. Stone, The generalized Weierstrass approximation theorem, Mathematics Magazine 21 (1948), 167–184.
- [3] G. Terzo, Some consequences of Schanuel's conjecture in exponential rings, Comm. Algebra 36 (2008), 1171–1189.

Ancillary file: `verify_theorem4.py` (mpmath 1.3.0, 50 decimal digits). All numerical claims verified by this script.